IMPERIAL COLLEGE LONDON

EE4-50
EE9-FPN2-10

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE PART IV: MEng and ACGI

SUSTAINABLE ELECTRICAL SYSTEMS

Thursday, 7 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	B. Chaudhuri, G. Strbac
	Second Marker(s) :	G. Strbac, B. Chaudhuri

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Answer ALL Questions

1. Question on Wind and Photovoltaic Power

a) The wind speeds before and after the plane of rotation of an ideal wind turbine are v_1 and v_2, respectively.

i) Show that the maximum power is captured by the wind turbine when v_2 is one-third of v_1. Make the simplifying assumptions as necessary.

[4]

ii) Use the above to calculate the theoretical maximum possible efficiency of power extraction from the wind.

[1]

b) Explain why variable speed operation is desirable for wind turbine generators.

[3]

c) A wind turbine with 80 m long blades has a maximum efficiency of 48% at the optimum tip-speed ratio of 7.5. Neglecting the losses in the generator and gearbox and assuming an air density of 1.22 kg/m^3, calculate the following:

i) the maximum power that can be generated (in MW) at a wind speed of 8 m/s.

[1]

ii) approximate mass flow rate (kg/s) of air through the wind turbine under the above condition in (i).

[2]

iii) the rotational speed of the turbine (in rpm) to capture the maximum power at a wind speed of 10 m/s.

[2]

d) A group of 24 solar PV modules are to be installed at a location with a latitude 35°N. Each module has a rated power of 300 W with 15% efficiency under standard test condition (STC). The terminal voltage of the PV module is 55 V for 300 W output under STC. A PV array is formed with 4 parallel strings with each string having 6 such PV modules in series with a string diode.

i) What is the approximate surface area (in m^2) of each PV module?

[1]

ii) What should be the tilt angle (in degrees with respect to the horizontal) of the PV modules at solar noon if the declination angle on a particular day is 20°?

[2]

iii) What is the maximum total power output of the PV array when the irradiance is 500 kW/m^2 assuming the same efficiency as under STC?

[2]

iv) How much power can be generated by the PV array under STC if two PV modules in separate strings are completely shaded?

[1]

v) How much power can be generated by the PV array under STC if two PV modules within the same string are completely shaded?

[1]

2. Question on balancing, Voltage Control and Security of Supply

Part A

a) Explain why large-scale penetration of wind generation will impose requirements for flexibility and explain why the need for "reserve" will increase.

[1]

b) Explain the advantages and disadvantages of providing reserve through "spinning" and "standing" reserve technologies.

[3]

Part B

c) Explain how on-load tap-changer voltage control could contribute to the amount of distributed generation that could be accommodated in a distribution network.

[3]

d) What will be the impact of active voltage control on network losses?

[1]

Part C

Contribution of distributed generation to the security of supply depends on the demand profiles and availability of the resources. In this case, the plant contains 3 units that can provide network support. Two of those are the same, each having rating of 10 MW and technical availability of each unit is 80%. Rating of the third unit is 20 MW with technical availability of 90%.

Demand profile during a peak season is represented by two characteristic days. Normalised load duration curves of these two characteristic days are shown in Figure 2.2.

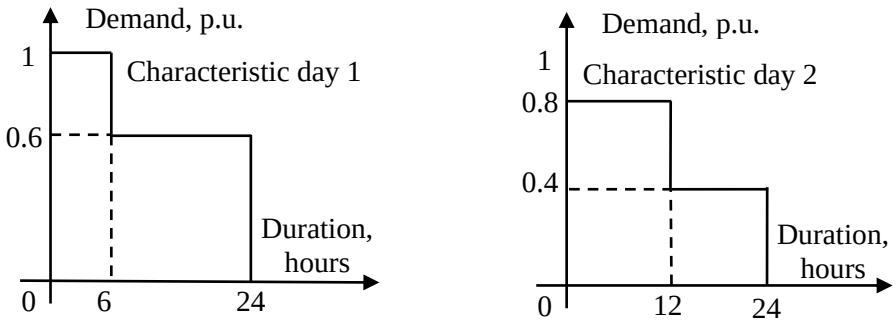


Figure 2.2 Normalised load duration curves for two characteristic days

During a peak season there are:

- 50 days for which load curve can be represented by using Characteristic day 1

- 50 days for which load curve can be represented by using Characteristic day 2

e) Sketch the peak season load duration curve that includes both peak day type, that will be used to calculate the security contribution of the distributed generation plant. When calculating the contribution to the security of supply, scale demand to the total generation plant capacity.

[2]

f) Form the capacity outage probability table for this plant, and calculate the corresponding probability values

[4]

g) Calculate the Expected Energy Not Supplied of the system supplied by the generation plant using the above load duration curve.

[3]

h) Calculate the capacity of the equivalent 100% reliable source (firm capacity), that results in the same Expected Energy Not Supplied.

[3]

3. Question on Low Carbon Heat

a) Give a brief description of the following types of low carbon heating systems.

i) Air source heat pump

[2]

ii) Hybrid heat pump

[2]

iii) District heating

[2.5]

iv) Hydrogen heating

[2.5]

b) In 2019 the UK introduced legislation to bring all greenhouse gas emissions to net-zero by 2050. This compares to its previous targets of at least 80% reduction from 1990 levels. In order to meet this target, it is likely that the removal of CO2 from the atmosphere will be required. Briefly explain why?

[2]

c) Two technologies for removing CO2 from the atmosphere are:

i) Bioenergy with carbon capture and storage (BECCS)

ii) Direct air capture of CO2 with storage (DACCS)

Give a brief description of each

[4]

d) A householder has decided to install a low carbon heating system and is trying to decide whether to convert their gas boiler to a hydrogen boiler or to purchase an air source heat pump (ASHP) system. The data are as follows:

	Heating system fixed costs	Running costs	Efficiency
Hydrogen boiler	£250pa	£0.18/kWh	90%
Air source heat pump	£1000pa	£0.15p/kWh	300%

(The fixed costs include the cost of the installation allocated over the life of the heating system.)

i) What is the annual heat demand whereby the total annual costs of the ASHP is the same as the H2 boiler?

[4]

ii) The householders' demand is 6,000kWhpa. Which system has the lowest annual cost?

[1]

4. Question on Flexibility, Hydro Power and Offshore Transmission

Demand Flexibility

A low-voltage feeder that is supplying a residential area, has a baseline (non-EV) demand profile peaking at 250 kW on a winter day, as shown in Figure 4.1. Due to favourable government incentives, 30 EVs have been purchased by the residential customers in the area. As specified in Table 4.1, they can be divided into 3 groups according to their driving patterns i.e. the time they arrive home and State of Charge (SoC) at arrival. Each EV has a 30 kWh battery, while the rating of each EV charger is 6 kW (for both charging and discharging, if allowed). Demand can be assumed constant in each 1-hour period. Ignore any charging or discharging losses. Answer the following questions:

a) If all EV users plug in their vehicles to charge immediately upon arrival and charge them at maximum capacity until their batteries are full, what is the total peak demand on the feeder and when does it occur? Plot the new total demand diagram.

[4]

b) By how many hours (at least) do all EVs need to delay their charging after arrival in order to avoid increasing the total peak demand of the feeder above the baseline peak demand of 250 kW? Assume that the same time delay for charging applies to all EVs.

[3]

c) Assume that EV Group No 1 is willing to provide V2G services (i.e. discharge the batteries) between 17:00 and 20:00. By how much can the total peak demand of the feeder be reduced? For the purpose of this question assume that all EVs across all groups can be charged after 3:00 am without any impact on the total peak demand of the feeder.

[3]

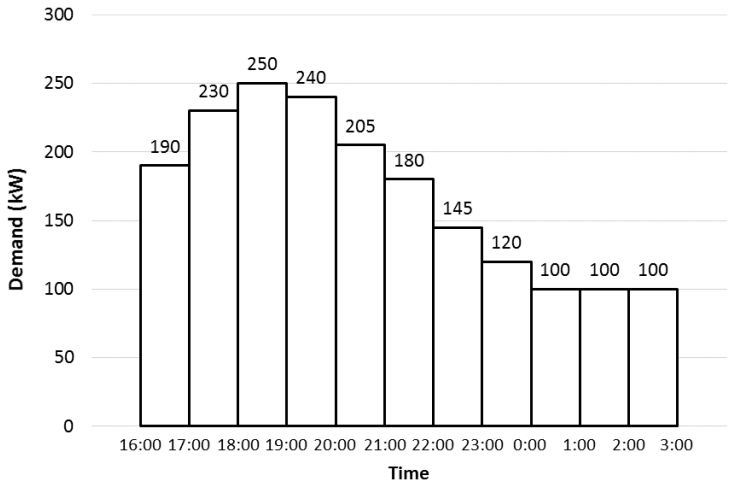


Figure 4.1. Distribution feeder baseline (non-EV) demand

Table 4.1. EV driving patterns and SoC across 3 vehicle groups

Group No.	Number of vehicles	Arrival time	SoC on arrival
1	5	17:00	80%
2	15	18:00	60%
3	10	19:00	40%

Hydro Power

d) A hydropower station with a reservoir has an effective head of 120 m. The height of the surface of the headwater in the reservoir is 150 m. There are 20 generators each rated at 700 MW, 20 kV, 50 Hz with a rated speed of 100 rpm.

i) What is the height (in m) of the surface of tailwater?

[1]

ii) How many poles are there in each generator?

[1]

iii) What type of generator rotor construction is necessary to support the poles and how is it different from the rotor of thermal generators?

[2]

iv) How much discharge (in m^3/s) of water is necessary to produce the rated power of the hydropower station assuming 90% efficiency of the generator.

[1]

Offshore Transmission

e) Explain why HVDC is preferred for high voltage subsea cables beyond a certain distance (i.e. cable length).

[3]

f) List two problems associated with the use of LCC HVDC for connecting remote offshore wind farms.

[2]